

HW SET-1

ANAGHA



Homework set 1

- a) Determine whether the following two logical propositions are logically equivalent:

$$(P \Leftrightarrow Q) \wedge P \quad \text{and} \quad P \Rightarrow Q.$$

- b) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by the expression $f(x) = x^2 - 3x + 2 - 3 \cdot |x - 1|$.

1. Determine whether the function is injective.
2. Determine the image set of the function.

- c) We are given the following two subsets of the complex numbers:

$$A = \{z \in \mathbb{C} \mid |z - i| = 2\} \quad \text{and} \quad B = \{z \in \mathbb{C} \mid \operatorname{Re}(z) = 1\}.$$

1. Draw the sets A and B in the complex plane.
2. Determine $A \cap B$.

- d) Show that the complex number $(1 - i)^{80}$ is a real number.

- e) As usual the principal value of the argument of a complex number z is denoted by $\operatorname{Arg}(z)$. Determine whether the following propositions are true:

1. $\operatorname{Arg}(z) = \pi \Rightarrow z \in \mathbb{R}_{>0}$.
2. $z \in \mathbb{R} \Rightarrow \operatorname{Arg}(z) = 0$.

The homework must be uploaded to the **DTU Learn** module for the course via "Assignments".
Deadline is **Sunday 28 September at 23:55**.

a) $(P \Leftrightarrow Q) \wedge P$ and $P \Rightarrow Q$

Corrections
↳ Red

P	Q	$P \Leftrightarrow Q$	$P \Leftrightarrow Q \wedge P$	$P \Rightarrow Q$
T	T	T	T	T
T	F	F	F	F
F	T	F	F	T
F	F	F	F	T

Since the truth tables are not identical, the two propositions are not logically equivalent.

b) $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = x^2 - 3x + 2 - 3 \cdot |x - 1|$$

1. Is the function injective?

Split into 2 cases



$$\begin{aligned} x - 1 &\geq 0 \\ x &\geq 1 \end{aligned}$$

$$\begin{aligned} x - 1 &< 0 \\ x &< 1 \end{aligned}$$

$$\begin{aligned} \text{Case 1: } & x^2 - 3x + 2 - 3(x - 1) \\ &= x^2 - 3x + 2 - 3x + 3 \\ &= x^2 - 6x + 5 \end{aligned}$$

$$\begin{aligned} \text{Case 2: } & x^2 - 3x + 2 - 3(1 - x) \\ &= x^2 - 3x + 2 - 3 + 3x \\ &= x^2 - 1 \end{aligned}$$

Hence,

$$f(x) = \begin{cases} x^2 - 1, & x < 1 \\ x^2 - 6x + 5, & x \geq 1 \end{cases}$$

Checking for monotonicity of each case:

→ $x^2 - 1$, parabola opening upwards

$$\text{derivative: } (x^2 - 1)' = 2x$$

$x < 0$, $2x < 0$, hence $f'(x) < 0$
 $f(x)$ decreasing

$x = 0$, $f'(0) = 0$
stationary point

If: $0 < x < 1$ then $2x > 0$

$f(x)$ $f'(x) > 0$
increasing

Hence, within first case, $f(x)$ is not monotone

Hence, function is not injective

2. image set of function

Case 1: $y \in \mathbb{R} \mid y \geq -1 \rightarrow [-1, \infty)$

Case 2: $(x^2 - 6x + 9) + 5 - 9 \rightarrow (x-3)^2 - 4$
 $y \in \mathbb{R} \mid y \geq -4 \rightarrow [-4, \infty)$

image set:

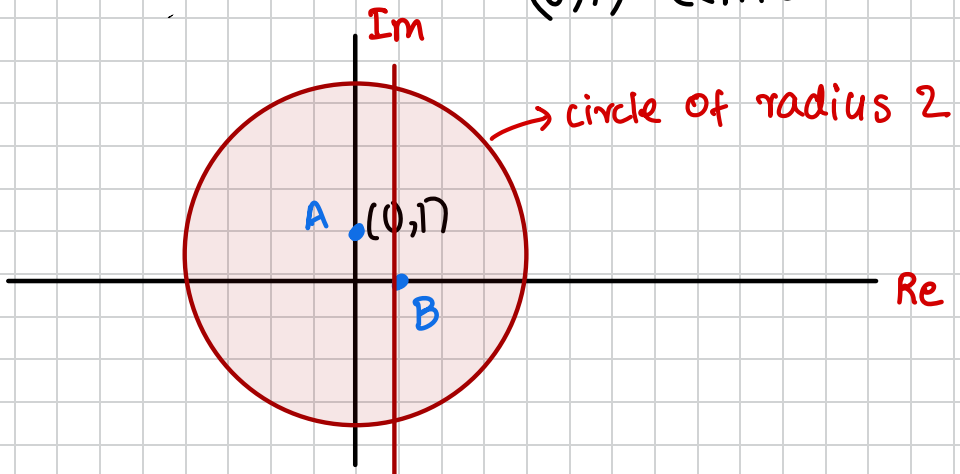
$$[-1, \infty) \cup [-4, \infty) \\ = [-4, \infty)$$

c) $A = \{z \in \mathbb{C} \mid |z - i| = 2\}$

$$B = \{z \in \mathbb{C} \mid \operatorname{Re}(z) = 1\}$$

$$\begin{aligned} |z - i| = 2 &= |x + iy - i| = 2 \\ &= \sqrt{x^2 + (y-1)^2} = 2 \\ &= x^2 + (y-1)^2 = 4 \\ &= \underbrace{(x-0)^2}_h + \underbrace{(y-1)^2}_k = 4 \end{aligned}$$

(0,1) center



2. $A \cap B$

$$A: x^2 + (y-1)^2 = 4$$

$$B: z = x + iy \quad \text{with } x = 1$$

$$\rightarrow x^2 + (y-1)^2 = 4$$

substituting $x=1$:

$$(1)^2 + (y-1)^2 = 4$$

$$(y-1)^2 = 3$$

$$y-1 = 1 \pm \sqrt{3} \rightarrow y = \underline{1 \pm \sqrt{3}}$$

since $\operatorname{Re}(z)=1$, solutions are:

$$1 + (1 + \sqrt{3}) \quad \& \quad 1 + (1 - \sqrt{3})$$

d) $(1-i)^{80}$
1-i in polar form

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \arctan\left(\frac{-1}{1}\right) = \frac{-\pi}{4}$$

$$\begin{aligned} \left(\sqrt{2} \operatorname{cis} \frac{-\pi}{4}\right)^{80} &= \left(\sqrt{2} \operatorname{cis} \frac{-\pi}{4}\right)^{80} \Rightarrow (\sqrt{2})^{80} \operatorname{cis} -20\pi \\ &= 2^{40} (\cos(-20\pi) + i \sin(-20\pi)) \\ &= 2^{40} (\cos(0) + i \sin(0)) \end{aligned}$$

$$\begin{aligned}
 &= 2^{40} (\cos(0) + i\sin(0)) \\
 &= 2^{40} (1 + 0) \\
 &= 2^{40}
 \end{aligned}$$

Therefore, $(1-i)^{80}$ is a real number

e) 1. $\text{Arg}(z) = \pi \Rightarrow z \in \mathbb{R}_{>0}$

If $\arg(z) = \pi$, complex number points along negative real axis, i.e. $\mathbb{R}_{<0}$

$$\text{as } z = re^{i\pi} = -r, r > 0$$

Hence, Proposition 1 is false

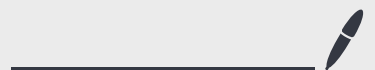
2. $z \in \mathbb{R} \Rightarrow \text{Arg}(z) = 0$

If $z \in \mathbb{R}$, there are two cases:

$$\begin{aligned}
 \rightarrow z > 0 &\Rightarrow z = r > 0 \rightarrow \text{Arg}(z) = 0 \\
 \rightarrow z < 0 &\Rightarrow z = -r < 0 \rightarrow \text{Arg}(z) = \pi
 \end{aligned}$$

Proposition 2 is false

HW SET
#2



Homework Assignment 2

- a) We are given complex numbers z_1 and z_2 . We are being informed that $\text{Arg}(z_1) = 2\pi/3$ and that $\text{Arg}(z_2) = 5\pi/6$. Show that $z_1 z_2$ is a purely imaginary number.
- b) Find two different solutions to the equation $e^z = 2 - 2i$.
- c) 1. Show by use of the division algorithm that the polynomial $Z^2 - 3Z + 2$ is a factor in the polynomial $2Z^4 - 6Z^3 + 8Z^2 - 12Z + 8$.
2. Now find all roots of the polynomial $2Z^4 - 6Z^3 + 8Z^2 - 12Z + 8$.
- d) 1. Show that the number -2 is a root of the polynomial $Z^4 + 4Z^3 + 5Z^2 + 4Z + 4$.
2. Determine the multiplicity of the root -2 of the polynomial $Z^4 + 4Z^3 + 5Z^2 + 4Z + 4$.
- e) We are given a real number $r \in \mathbb{R}_{>0}$. Show using induction on n that

$$(1 + r)^n \geq 1 + nr$$

for all $n \in \mathbb{Z}_{\geq 1}$.

Your solution must be uploaded to the **DTU Learn** module for the course under "Assignments".
Deadline is **Sunday October 26 at 23:55**.

$$a) \arg(z_1) = \frac{2\pi}{3} \quad \text{let } \text{mod}(z_1) \text{ be } r_1$$

$$z_1 = r_1 e^{\frac{2\pi}{3}}$$

$$\arg(z_2) = \frac{5\pi}{6} \quad \text{let } \text{mod}(z_2) \text{ be } r_2$$

$$z_2 = r_2 e^{\frac{5\pi}{6}}$$

$$z_1 \cdot z_2 = r_1 e^{\frac{2\pi}{3}} \cdot r_2 e^{\frac{5\pi}{6}}$$

$$= r_1 \cdot r_2 \cdot e^{\frac{2\pi}{3} + \frac{5\pi}{6}}$$

$$= r_1 \cdot r_2 \cdot e^{\frac{3\pi}{2}}$$

$$= r_1 \cdot r_2 e^{\frac{3\pi}{2}}$$

$$= r_1 \cdot r_2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right)$$

$$= r_1 \cdot r_2 (0 - i)$$

$$= -i (r_1 \cdot r_2)$$

Hence, the complex Number is purely imaginary

$$b) e^z = 2 - 2i$$

$$z = a + bi$$

$$e^{a+bi} = 2 - 2i$$

$$e^a \cdot e^{bi} = 2 - 2i$$

$2 - 2i$ in Polar form:

$$r = \sqrt{(2)^2 + (-2)^2} = 2\sqrt{2}$$

$$\arg(2 - 2i) = \frac{-2}{2} = -1 \rightarrow -\frac{\pi}{4}$$

$$2 - 2i = 2\sqrt{2} e^{-\pi/4 i}$$

$$e^a \cdot e^{bi} = 2\sqrt{2} e^{-\pi/4 i}$$

same modulus:

$$e^a = 2\sqrt{2}$$

$$a = \ln 2\sqrt{2}$$

same argument

$$b = \left(-\frac{\pi}{4} + 2\pi p \right) i, \\ p \in \mathbb{Z}$$

$$z = \ln 2\sqrt{2} + \left(-\frac{\pi}{4} + 2\pi p \right) i$$

$$z_1 = \ln 2\sqrt{2} - \frac{\pi}{4} i$$

$$z_2 = \ln 2\sqrt{2} + \frac{7\pi}{4} i$$

c) 1. $z^2 - 3z + 2$ is a factor
of $2z^4 - 6z^3 + 8z^2 - 12z + 8$

$$\begin{array}{r}
 z^2 - 3z + 2 \mid 2z^4 - 6z^3 + 8z^2 - 12z + 8 \quad | 2z^2 + 4 \\
 \underline{2z^4 - 6z^3 + 4z^2} \\
 (-) \quad (+) \quad (-) \\
 4z^2 - 12z + 8 \\
 \underline{4z^2 - 12z + 8} \\
 (+) \quad (+) \quad (-) \\
 0
 \end{array}$$

Since remainder = 0, $z^2 - 3z + 2$
is a factor of $2z^4 - 6z^3 + 8z^2 - 12z + 8$

2. $(z^2 - 3z + 2)(2z^2 + 4) = 2z^4 - 6z^3 + 8z^2 - 12z + 8$
 $\rightarrow z^2 - 3z + 2 = 0$

$$\begin{aligned}
 \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(2)}}{2} &= \frac{3 \pm \sqrt{1}}{2} \\
 &= \frac{3 \pm 1}{2} \begin{cases} \rightarrow \frac{3+1}{2} = 2 \\ \rightarrow \frac{3-1}{2} = 1 \end{cases}
 \end{aligned}$$

Hence, factors $\rightarrow (z-1)(z-2)$

$$\begin{aligned}
 \rightarrow 2z^2 + 4 &= 0 \\
 z^2 + 2 &= 0 \\
 z^2 &= -2 \\
 z &= \pm \sqrt{-2} \\
 &= \pm i\sqrt{2}
 \end{aligned}$$

Hence, factors are:

$$\text{Therefore, roots} = i\sqrt{2}, -i\sqrt{2}, 1, 2$$

d) 1. $z - (-2)$ is a factor of $z^4 + 4z^3 + 5z^2 + 4z + 4$

$$\begin{array}{r} z+2 \overline{) z^4 + 4z^3 + 5z^2 + 4z + 4} \quad | \quad z^3 + 2z^2 + z + 2 \\ \underline{z^4 + 2z^3} \\ 2z^3 + 5z^2 \\ \underline{2z^3 + 4z^2} \\ z^2 + 4z \\ \underline{z^2 + 2z} \\ 2z + 4 \\ \underline{2z + 4} \\ 0 \end{array}$$

Since the remainder is zero, (-2) is a factor of the polynomial

2. Multiplicity of root.

$$(z+2)(z^3 + 2z^2 + z + 2) = z^4 + 4z^3 + 5z^2 + 4z + 4$$

$$z^3 + 2z^2 + z + 2 \div z + 2$$

P.T.O

$$\begin{array}{r}
 z+2 \mid z^3 + 2z^2 + z + 2 \mid z^2 + 1 \\
 \underline{z^3 + 2z^2} \\
 z + 2 \\
 \underline{z + 2} \\
 0
 \end{array}$$

Hence.
$$z^3 + 2z^2 + z + 2 = (z+2)(z^2+1) + 0$$

$$\begin{aligned}
 z^4 + 4z^3 + 5z^2 + 4z + 4 &= (z+2)(z^3 + 2z^2 + z + 2) \\
 &= (z+2)(z+2)(z^2+1) \\
 &= (z+2)^2(z^2+1)
 \end{aligned}$$

Hence, multiplicity of root $-2 = 2$

e) $P(n): (1+r)^n \geq 1+nr$ for all $n \in \mathbb{Z}_{\geq 1}$

Base case: $P(1)$

$$(1+r)^1 \geq 1+1 \cdot r$$

$$1+r = 1+r$$

Hence, $P(1)$ is true

P.T.O

Inductive Hypothesis:

Assume $P(n-1)$ to be true

$$P(n-1): (1+r)^{n-1} \geq 1 + (n-1)r$$

$$\frac{(1+r)^n}{1+r} \geq 1 + rn - r$$

Multiplying both sides with $1+r$

$$(1+r)^n \geq (1+rn-r)(1+r)$$

$$(1+r)^n \geq 1+rn-r+r+r^2n-r^2$$

$$(1+r)^n \geq 1+rn+r^2(n-1)$$

Since $r > 0$ and $n-1 \geq 0$
because $n \geq 1$

$$(n-1)r^2 \geq 0$$

$$\text{Hence, } 1+rn+r^2(n-1) \geq 1+rn$$

$$(1+r)^n \geq 1+rn$$

$$(1+r)^n \geq 1+rn+r^2(n-1) \geq 1+rn$$

If $P(n-1)$ is true, $P(n)$ is also true.

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{Z}_{\geq 1}$

HW SET

3

ANAGHA 

Homework Assignment 3

Solve the following exercises without use of electronic aids.

- a) Let W be spanned by the following vectors in $\mathbb{R}^3$:

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}, \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} 7 \\ 11 \\ 6 \end{bmatrix}.$$

Provide an ordered basis for W .

- b) Let $C_\infty(\mathbb{R})$ be the real vector space from Example 10.4.5 in the textbook. We define a function $L : C_\infty(\mathbb{R}) \rightarrow C_\infty(\mathbb{R})$ by $L(f) = f' + f - 1$, where the expression f' denotes the derivative function of f . Is L a linear map?

- c) Let $F : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be defined as follows:

$$F\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 \\ -4 & 5 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad v_1, v_2 \in \mathbb{C}.$$

We are given the ordered bases:

$$\beta = \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix}\right) \quad \text{and} \quad \gamma = \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \quad \text{for } \mathbb{C}^2.$$

Calculate the mapping matrix ${}_\beta[F]_\gamma$.

- d) We choose the following ordered basis for the real vector space $\mathbb{R}^{2 \times 2}$:

$$\beta = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right).$$

We are given the linear map $M : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ defined by

$$M(\mathbf{A}) = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \cdot \mathbf{A}, \quad \mathbf{A} \in \mathbb{R}^{2 \times 2}.$$

Compute the mapping matrix ${}_\beta[M]_\beta$.

- e) We are given the following matrix:

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

Determine the eigenvalues of the matrix as well as ordered bases for their corresponding eigenspaces.

- f) We are being informed that

$$\mathbf{v}_p = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in \mathbb{R}^2$$

is a particular solution to an inhomogeneous system of linear equations over $\mathbb{R}$ with four equations and two unknowns. Is the vector $3 \cdot \mathbf{v}_p$ a solution to the system?

- g) Let V be the real vector space $\mathbb{R}^{3 \times 3}$. State a subspace of V with a dimension of 5 and justify your answer.

Your solution must be uploaded to the course's **DTU Learn** module under "Assignments". The deadline is **Sunday November 23 at 23:55**.

$$a) \quad v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 7 \\ 11 \\ 6 \end{bmatrix}$$

RREF(v_1, v_2, v_3)

$$\begin{bmatrix} -1 & 5 & 7 \\ 1 & 4 & 11 \\ 0 & 3 & 6 \end{bmatrix} \xrightarrow{R_3/3} \begin{bmatrix} -1 & 5 & 7 \\ 1 & 4 & 11 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_1 + R_2} \begin{bmatrix} -1 & 5 & 7 \\ 0 & 9 & 18 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\downarrow R_2/9$$

$$\begin{bmatrix} 1 & -5 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xleftarrow{R_3 \leftarrow R_2 - R_3} \begin{bmatrix} 1 & -5 & 7 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

Pivots are in first and second columns

Hence, ordered basis for $W = (v_1, v_2)$

$$W = \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix} \right)$$

$$b) \quad L: C_\infty(\mathbb{R}) \rightarrow C_\infty(\mathbb{R})$$

$$L(f) = f' - f - 1 \quad \text{Is } L \text{ a linear map?}$$

$$L(f_1) = f_1' + f_1 - 1$$

$$L(f_2) = f_2' + f_2 - 1. \quad \text{Let } c \text{ be any scalar}$$

$$L(f_1) + c L(f_2) = f_1' + f_1 - 1 + c f_2' + c f_2 - c$$

$$L(f_1 + c \cdot f_2) = f_1' + c \cdot f_2' + f_1 + c \cdot f_2 - 1$$

Since $L(f_1) + c \cdot L(f_2) \neq L(f_1 + cf_2)$, the map L is not linear

$$c) {}_{\beta}[F]_{\gamma} = \left((F \begin{bmatrix} 1 \\ 1 \end{bmatrix})_{\beta}, (F \begin{bmatrix} 0 \\ 1 \end{bmatrix})_{\beta} \right)$$

$$F \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\left(F \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)_{\beta} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{matrix} c_1 - 2c_2 \\ 2c_1 + c_2 \end{matrix} \Rightarrow \begin{matrix} c_1 - 2c_2 = 2 \\ (2c_1 + c_2 = 1) \times 2 \end{matrix}$$

$$\begin{aligned} &\Rightarrow \begin{matrix} c_1 - 2c_2 = 2 \\ 4c_1 + 2c_2 = 2 \end{matrix} \\ &+ \hline &\quad \underline{5c_1} = 4 \\ &\quad c_1 = \frac{4}{5} \end{aligned}$$

$$c_2 = \frac{c_1 - 2}{2} = \frac{1}{2} \left(\frac{4}{5} - 2 \right) = \frac{1}{2} \cdot \frac{-6}{5} = \frac{-3}{5}$$

$$\text{Hence, } \left(F \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)_{\beta} = \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$$

$$F \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$\left(F \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)_{\beta} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 5 \end{bmatrix} \begin{cases} c_1 - 2c_2 \\ 2c_1 + c_2 \end{cases} \Rightarrow \begin{cases} c_1 - 2c_2 = 1 \\ (2c_1 + c_2 = 5) \times 2 \end{cases}$$

$$\Rightarrow \begin{array}{r} c_1 - 2c_2 = 1 \\ 4c_1 + 2c_2 = 10 \\ \hline 5c_1 = 11 \end{array}$$

$$c_1 = \frac{11}{5}$$

$$c_2 = \frac{c_1 - 1}{2} = \frac{1}{2} \left(\frac{11}{5} - 1 \right) = \frac{1}{2} \cdot \frac{6}{5} = \frac{3}{5}$$

$$\left[F \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right]_{\beta} = \begin{bmatrix} 11/5 \\ 3/5 \end{bmatrix}$$

$$\text{Hence } {}_{\beta}[F]_{\gamma} = \begin{bmatrix} \frac{4}{5} & \frac{11}{5} \\ -\frac{3}{5} & \frac{3}{5} \end{bmatrix}$$

$$d) {}_{\beta}[M]_{\beta} = \left((M \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix})_{\beta}, (M \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix})_{\beta}, (M \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix})_{\beta}, (M \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix})_{\beta} \right)$$

$$M \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\left(M \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right)_{\beta} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$M \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ -2 & 0 \end{bmatrix}$$

$$\left[M \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right]_{\beta} = \begin{bmatrix} 2 \\ -2 \\ 0 \\ 0 \end{bmatrix}$$

$$M \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

$$\left[M \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right]_{\beta} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$M \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix}$$

$$\left[M \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right]_{\beta} = \begin{bmatrix} 0 \\ 0 \\ 2 \\ -2 \end{bmatrix}$$

$$\text{Hence } {}_{\beta} M_{\beta} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -1 & -2 \end{bmatrix}$$

$$2) \begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 1-\lambda & -1 \\ 2 & -1 & 1-\lambda \end{bmatrix}$$

$A \qquad \qquad \qquad I\lambda$

$$\det(A - I\lambda) = 2 - \lambda ((1 - \lambda)(1 - \lambda) - (-1)(-1)) - 0 + 0$$

$$\begin{aligned} \det(A - I\lambda) &= (2 - \lambda)((1 - \lambda)^2 - 1) \\ &= (2 - \lambda)(1 + \lambda^2 - 2\lambda - 1) \\ &= (2 - \lambda)(\lambda^2 - 2\lambda) \\ &= (2 - \lambda)(\lambda)(\lambda - 2) \end{aligned}$$

Equating $\det(A - I\lambda)$ to 0

$$0 = (2 - \lambda)(\lambda)(\lambda - 2)$$

$$\boxed{\lambda = 0, \lambda = 2}$$

To find eigenspaces

$$(A - I\lambda)x = 0$$

$$\lambda = 0, (A)x = 0$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_2 + R_3} \begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & -1 \\ 4 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{\substack{R_1/2 \\ R_3/4}} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_1 - R_3} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 = 0$$

$$x_1 = 0$$

$$-x_2 + x_3 = 0$$

$$x_2 - x_3 = 0$$

$$x_2 = x_3$$

$$x_3 = t$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} t$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} t$$

basis for eigenspace: $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$$\lambda = 2, (A - 2I) = 0$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 2 & -1 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

RREF:

$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & -1 & -1 \\ 2 & -1 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & -1 & -1 \\ 2 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_1 - R_2} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 - x_2 - x_3 = 0$$

$$2x_1 = x_2 + x_3$$

$$x_1 = \frac{x_2}{2} + \frac{x_3}{2}$$

$$\text{Let } x_2 = s \quad \& \quad x_3 = t$$

$$\text{Hence } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = s \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \\ 1 \end{bmatrix} t$$

$$\text{Hence ordered basis: } \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 0 \\ 1 \end{bmatrix}$$

$$f) v_p = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in \mathbb{R}^2$$

Inhomogeneous system of linear equations can be written as:

$$Ax = b$$

$$\text{If } x = 3v_p$$

$$A(3v_p) = b$$

for $3v_p$ to be a solution, it must satisfy $A(3v_p) = b$

This requires $3b = b$, which only holds if $b = 0$

However, since this system is inhomogeneous, it is not possible for b to be equal to 0.

Hence $3b \neq b$

Hence $3 \cdot v_p$ is not a solution

$$g) V = \mathbb{R}^{3 \times 3}$$

$$\dim(V) = 3 \times 3 = 9$$

Consider the set:

$$U = \left\{ \begin{pmatrix} a & b & c \\ d & e & 0 \\ 0 & 0 & 0 \end{pmatrix} : a, b, c, d, e \in \mathbb{R} \right\}$$

$$\text{Let } X = \begin{pmatrix} a_1 & b_1 & c_1 \\ d_1 & e_1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} a_2 & b_2 & c_2 \\ d_2 & e_2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Checking X & Y for closure under addition & scalar multiplication.

$$X + cY \quad \text{where } c \in \mathbb{R} \\ X, Y \in U$$

$$\Rightarrow \begin{pmatrix} a_1 & b_1 & c_1 \\ d_1 & e_1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + c \begin{pmatrix} a_2 & b_2 & c_2 \\ d_2 & e_2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} a_1 + ca_2 & b_1 + cb_2 & c_1 + cc_2 \\ d_1 + cd_2 & e_1 + ce_2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in U$$

Hence, U is a subspace for V

$$\text{Basis for } U: \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\}$$

$$\dim(U) = 5$$

For Today:

☒ Past Exams

☒ Exercises

☒ Möbius

☒ Hw sets

☒ Extra Exercises

→ PAST EXAMS ←

$$\begin{array}{ccc|c} 0 & 1 & 1 & 1 \\ 1 & -2 & 1 & 4 \end{array}$$

$$\begin{array}{ccc|c} 1 & -2 & 1 & 4 \\ 0 & 1 & 1 & 1 \end{array} \rightarrow \begin{array}{ccc|c} 1 & 0 & 3 & 6 \\ 0 & 1 & 1 & 1 \end{array}$$

$$\begin{aligned} x_1 &= 6 - 3x_3 \\ x_2 &= 1 - x_3 \end{aligned} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \\ 1 \end{bmatrix} t$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 & -1 \\ 11 & 4 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 11 \\ 2 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

A matrix is invertible if $\det \neq 0$

$$(-1)(-1) \begin{bmatrix} 1 & 9 \\ 1 & 1 \end{bmatrix} + a \begin{bmatrix} 1 & 2 \\ 1 & a \end{bmatrix}$$

$$(1-9) + a[a-2]$$

$$-8 + a^2 - 2a$$

$$a^2 - 2a - 8$$

$$a^2 - 4a + 2a - 8$$

$$a(a-4) + 2(a-4)$$

$$(a-4)(a+2)$$

$$a \in \mathbb{R} \setminus \{-2, 4\}$$

$$\begin{array}{r|l} z+1 & z^3 - 5z^2 - z + 5 \\ \hline & z^3 + z^2 \\ & \hline & -6z^2 - z \\ & -6z^2 - 6z \\ & \hline & 5z + 5 \\ & 5z + 5 \\ & \hline & 0 \end{array} \quad \begin{array}{l} z^2 - 6z + 5 \\ (z-5)(z-1) \end{array}$$

eigenvalues: $\begin{bmatrix} 4-\lambda & -1 & -1 \\ 6 & -3-\lambda & -1 \\ -6 & 5 & 3-\lambda \end{bmatrix}$

If λ is an eigenvalue

$$\begin{aligned} \begin{bmatrix} 0 & -1 & -1 \\ 6 & -7 & -1 \\ -6 & 5 & -1 \end{bmatrix} &\rightarrow \begin{bmatrix} 0 & -1 & -1 \\ 6 & -7 & -1 \\ 0 & -2 & -2 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 6 & -7 & -1 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & -7 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 6 & 0 & -6 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \end{aligned}$$

$f: v \rightarrow v$

$$\begin{bmatrix} 1-\lambda & 1 & 0 \\ 1 & -1-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

$$(1-\lambda) \begin{bmatrix} (1-\lambda) & (-1-\lambda) & -1 \end{bmatrix}$$

$$(1-\lambda) (-1-\cancel{\lambda} + \cancel{\lambda} + \lambda^2 - 1)$$

$$(1-\lambda)(\lambda^2 - 2)$$

$$\lambda = \pm \sqrt{2}, 1$$

$$\begin{array}{cc} -1-\lambda & 3 \\ 3 & -1-\lambda \end{array} \quad (1-\lambda)^2 - 9$$

$$1 + \lambda^2 + 2\lambda - 9$$

$$\lambda^2 + 2\lambda - 8$$

$$(\lambda + 4)(\lambda - 2)$$

$$\lambda = -4 \quad \lambda = +2$$

$$\begin{bmatrix} -1-2 & 3 \\ 3 & -1-2 \end{bmatrix} \rightarrow \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-4t} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

$$c_1 = -c_2$$

$$c_1 - c_2 = 6$$

$$-2c_2 = 6$$

$$c_2 = -3$$

$$c_1 = 3$$

$$\begin{aligned} (z-2i)(z-1-i) &= z^2 - z - iz - 2iz + 2i + 2i^2 \\ &= z^2 + z(-1-i-2i) + 2i - 2 \\ &= z^2 + z(-3i-1) + (2i-2) \end{aligned}$$

P	Q	$P \vee Q$	$\neg(P \vee Q)$	$\neg(P \vee Q) \Rightarrow P$
T	T	T	F	T
F	F	F	T	F
T	F	T	F	T
F	T	T	F	T



Logically equivalent

$$z^3 + 27$$

$$z^3 = -27$$

$$\text{let } z = re^{i\theta}$$

$$(re^{i\theta})^3 = -27 + 0i$$

SAME MOD:

$$r^3 = |-27|$$

$$r = 3$$

SAME ARGUMENT:

$$3\theta = \arg\left(-\frac{27}{0}\right)$$

$$3\theta = \pi + 2p\pi$$

$$\theta = \frac{\pi}{3} + \frac{2p\pi}{3}$$

Hence, roots:

$$3e^{i\frac{\pi}{3}}$$

$$3e^{i\pi}$$

$$3e^{i\frac{5\pi}{3}}$$

$$\pi + 4\pi$$

$$3\left(\frac{1}{2} + \frac{j\sqrt{3}}{2}\right)$$

$$3(-1)$$

$$3\left(\frac{1}{2} - \frac{j\sqrt{3}}{2}\right)$$

$$\frac{3}{2} + \frac{j\sqrt{3}}{2}$$

$$-3$$

$$\frac{3}{2} - \frac{j\sqrt{3}}{2}$$

a) $\underbrace{\quad}_{\underline{A}} \quad \underbrace{\quad}_{\underline{b}}$

$\underline{b} \neq 0$ system is inhom

b) If $\underline{v} \in \mathbb{R}^4$ & $\underline{w} \in \mathbb{R}^4$ are 2 different solutions to inhom,

$$\underline{A} \underline{v} = \underline{b} \quad \underline{A} \underline{w} = \underline{b}$$

$$\underline{A} \underline{v} = \underline{A} \underline{w} = \underline{b}$$

$$\underline{A} (\underline{v} - \underline{w}) = \underline{A} \underline{v} - \underline{A} \underline{w} = 0$$

solution to hom but not

$$\begin{bmatrix} 1-\lambda & 0 & 0 \\ 4 & -\lambda & 1 \\ 3 & 0 & 2 \end{bmatrix} \xrightarrow{+1} (-1)(1-\lambda) \begin{bmatrix} -\lambda & 1 \\ 0 & 8-\lambda \end{bmatrix} \xrightarrow{\text{inhom}} -0 \neq 0$$

$$= (1-\lambda) \begin{bmatrix} -\lambda & 1 \\ 0 & 8-\lambda \end{bmatrix} = (1-\lambda) (-1)(8-\lambda)$$

$$(-\lambda + \lambda^2)(8-\lambda)$$

$$-8\lambda + \lambda^2 + 8\lambda^2 - \lambda^3$$

$$-\lambda^3 + 9\lambda^2 - 8\lambda$$

eigenvalues are $\lambda=0$, $\lambda=1$, $\lambda=8$

Consider $\underline{A}$ as a mapping matrix of a map. $\mathbb{R}^3 \rightarrow \mathbb{R}^3$

We map the eigenvectors -28+3

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 0 & 1 \\ 3 & 0 & 8 \end{bmatrix} \begin{bmatrix} -7 \\ -25 \\ 3 \end{bmatrix} = \begin{bmatrix} -7 \\ -25 \\ 3 \end{bmatrix}$$

eigenvalue = 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 0 & 1 \\ 3 & 0 & 8 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 64 \end{bmatrix}$$

eigenvalue $\lambda_2 = 8$

$$\rightarrow \left[\begin{array}{cc|cc} 1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ 1 & -1 & 1 & 0 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{array} \right]$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\left(t + \frac{3}{2} \right)'' - 3 \left(t + \frac{3}{2} \right)' + 2 \left(t + \frac{3}{2} \right)$$

$$0 - \cancel{3} + 2t + \cancel{3}$$

$$= 2t$$

$$r^2 - 3r + 2r = 2t$$

$$\Delta = \text{discr: } 9 - 4(2) = 1$$

$$\text{roots: } \frac{-(-3) \pm \sqrt{1}}{2} = \frac{3 \pm 1}{2} = 2, 1$$

$$\text{hence, sol: } C_1 e^{2t} + C_2 e^t + \frac{3}{2} t$$

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 3 & 0 \\ 0 & 7 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -2 & 9 \\ 1 & 6 & 0 \\ 0 & 14 & -6 \end{bmatrix}$$

$$-14 \begin{bmatrix} 2 & 9 \\ 1 & 0 \end{bmatrix} - 6 \begin{bmatrix} 2 & -2 \\ 1 & 0 \end{bmatrix}$$

$$-14(-9) - 6(12 + 2)$$

$$-14(-9) - 6(14)$$

$$14(9 - 6) = 14(3) = 42$$

$$x = A^{-1} b$$

$$\frac{1}{2} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 12 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 12 \end{bmatrix} \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

$$36 - 4(6)(3)$$

$$36 - 72$$

$$-36 \rightarrow -36$$

$$\frac{6 \pm 6i}{6}$$

$$1 \pm i$$

$$2t = t + t$$

$$r^2 - 6r + 9$$

$$36 - 4(9) = 0$$

$$\frac{6}{2} = 3$$

$$c_1 e^{3t} + c_2 t e^{3t}$$

$$2 \begin{bmatrix} 2-\lambda & 5 \\ 0 & -2 \end{bmatrix} + (-\lambda) \begin{bmatrix} 2-\lambda & 3 \\ 0 & -\lambda \end{bmatrix}$$

$$\begin{aligned} & (-2)(2)(2-\lambda) + (-\lambda)(2-\lambda)(-\lambda) \\ & (2-\lambda)(-4 + \lambda^2) \end{aligned}$$

$$\lambda^2 = 4 \quad \lambda = 2$$

$$\lambda = \pm 2$$

$$\begin{bmatrix} 0 & 3 & 5 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 3 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 & 5 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 3 & 5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 0 & 8 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\underline{A} = V D V^{-1}$$

$$= \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}^{-1}$$

P	Q	R	$P \Rightarrow R$	$(P \Rightarrow R) \Leftrightarrow Q$	$R \Rightarrow P$	*
T	T	T	T	T	T	T
F	T	T	T	F	F	F
T	F	T	T	F	F	F
F	F	T	T	F	F	T
T	T	F	F	F	T	T
F	T	F	T	T	T	T
T	F	F	F	F	T	F
F	F	F	T	F	T	F

$$(e^{i\pi/2})(2+i) \rightarrow (i)(2+i)$$

$$\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)^4 \rightarrow \sqrt{\frac{2}{4} + \frac{2}{4}} = \sqrt{1} = 1$$

$$\text{argument} = \tan^{-1}\left(\frac{\sqrt{2}/2}{\sqrt{2}/2}\right) = \frac{\pi}{4}$$

$$\left(\frac{\sqrt{2}}{2} e^{i\pi/4}\right)^4 = 1 e^{i\pi/4 \cdot 4}$$

$$= e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1$$

$$\det \begin{bmatrix} 1 & -2 \\ -3 & 7 \end{bmatrix} \rightarrow 7 - 6 = 1$$

$$\det \neq 0$$

hence it is invertible

$B \in \mathbb{R}^{n \times n}$ an invertible matrix.

If zero is an eigenvalue of B .

So a vector $\underline{v}$, exists such that

$$\underline{B} \cdot \underline{v} = 0 \quad \underline{v} \neq 0$$

where $\underline{v} \neq 0$

if $\underline{B} \cdot \underline{v} = 0$, then the Kernel of the matrix is not trivial

The matrix would then not have full rank.

However, an inverse exists only if a matrix is full rank.

Hence, not possible

$$a) 1+2z, 2+z-z^2 \text{ \& } z^2$$

$$[\gamma_1]_\gamma = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad [\gamma_2]_\gamma = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad [\gamma_3]_\gamma = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{basis for kernel: } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Since γ_1 & γ_2 map to the zero polynomial, hence, they belong to kernel however γ_3 doesn't map to 0.

$$[1+2z+z^2]_\gamma = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \notin \text{ker}$$

image(L) = colspace of the mapping matrix.

$$\text{basis for column space is } w_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}_\gamma = 0(\gamma_1) + 1(\gamma_2) + 0(\gamma_3) = 2+z-z^2$$

$$L(p(z)) = \gamma L_\gamma \cdot [V]_\gamma$$

$$2 - z + 3z^2 = c_1(1 - z) + c_2(1 + z^2) + c_3(z + z^2)$$

$$4 - z - 3z^2 = c_1 - c_1z + c_2 + \underline{c_2z^2} + c_3z + \underline{c_3z^2}$$

$$c_3 - c_1 = -1 \rightarrow c_1 = c_3 + 1$$

$$c_1 + c_2 = 4 \rightarrow c_2 = 4 - c_1$$

$$c_3 + c_2 = 3$$

$$c_3 + 4 - c_1 = 3$$

$$c_3 - c_1 = -1$$

$$1 - z, z + 2z^2, 3 - z + z^2$$

$$\text{Basis for kernel: } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$[\gamma_1]_\gamma = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \gamma_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \gamma_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$[\gamma_1 \text{ \& } \gamma_3] \in \ker$$

$$2 - z + 3z^2 \rightarrow p(z) \notin \ker$$

homogenous

eigenvalues: $\begin{matrix} 2-\lambda & 1 \\ 5 & -2-\lambda \end{matrix}$

$$(2-\lambda)(-2-\lambda) - 5$$

$$-4 - 2\lambda + 2\lambda + \lambda^2 - 5$$

$$\lambda^2 - 9 = 0$$

$$\lambda = \pm 3$$

eigenspaces:

$$\begin{matrix} & 5 - (-9) \\ \begin{bmatrix} -1 & 1 \\ 5 & -5 \end{bmatrix} \end{matrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\xrightarrow{-2+3} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 1 \\ 5 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1/5 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1/5 \\ 0 \end{bmatrix}$$

$$c_1 e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} -1/5 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ -1 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix}$$

$$B = A^{-1}C$$

$$8-12$$

$$\frac{1}{-2} \begin{bmatrix} -2 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & -4 \\ -4 & 6 \end{bmatrix}$$

$$\frac{1}{-2} \begin{bmatrix} 0 & -4 \\ -4 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 2 \\ 2 & -3 \end{bmatrix}$$

$$-K \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} - 3 \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$$

$$-K(2) - 3(1-2) = 0$$

$$-2K - 3 = 0$$

$$2K + 3 = 0$$

$$K = \frac{-3}{2}$$

$$p(b) = 10$$

$$c = 10$$

$$(1-2i)(1+2i) = \frac{10}{a}$$

$$1+4 = \frac{10}{a}$$

$$a = 2$$

$$\begin{array}{r} -4+1+9 \\ -10-6 \end{array} \quad \begin{array}{r} 6 \\ -16 \end{array}$$

$$\text{sum: } 2 = \frac{-(-6)}{2}$$

$$4 = b$$

$$\begin{array}{r} 0 \\ -1/2 \\ 1 \end{array} \rightarrow \begin{array}{r} 0 \\ -1 \\ 2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & 8 & 4 \end{bmatrix} \quad \begin{array}{l} 1-2(2) \\ x_2=2 \end{array}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

P	Q	R	$\neg P$	$\neg P \wedge Q$	$P \vee Q$	$\star$
T	T	T	F	F	T	T
F	T	T	T	T	T	T
T	F	T	F	F	T	T
F	F	T	T	F	F	T
T	T	F	F	F	T	T
F	T	F	T	T	T	T
T	F	F	F	F	T	T
F	F	F	T	T	F	T

$$z^4 = -4$$

$$(re^{i\theta})^4 = -4$$

$$r^4 = \sqrt{-4}^2 \text{ SAME MOD} = \sqrt{2}$$

$$r = \sqrt{2}$$

SAME ARG

$$4\theta = \arg(-4)$$

$$4\theta = \pi + 2p\pi$$

$$\theta = \frac{\pi}{4} + \frac{p\pi}{2}$$

in first Quadrant.

$$\begin{aligned} z &= 2e^{i\pi/4} = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\ &= \sqrt{2} \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \\ &= \frac{1}{2} + \frac{i}{2} \end{aligned}$$

$$v_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 2 \\ -3 & 1 & 7 \\ 2 & 1 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & 13 & 13 \\ 0 & 7 & 7 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\{v_1, v_2\}$

$\mathbb{R}^3$ that doesn't lie in $\mathbb{R}^2$

$$\begin{bmatrix} 1 & 4 & 1 \\ -3 & 1 & 3 \\ 2 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 1 \\ 0 & 13 & 6 \\ 0 & 7 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 4 & 1 \\ 0 & 1 & 0 \\ 0 & 13 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 13 & 0 & -11 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}_e[L]_e = {}_e \text{id}_\gamma \gamma [L]_\gamma \gamma \text{id}_e$$

$$2^{-\frac{1}{2}}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \gamma_2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \gamma_2 & 0 \\ 0 & 2 & 0 & 1 & -\gamma_2 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \gamma_2 & 0 \\ 0 & 1 & 0 & \gamma_2 & -\gamma_4 & 0 \\ 0 & 0 & 1 & -\gamma_2 & \gamma_4 & 1 \end{array} \right]$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{4} & 0 \\ -\frac{1}{2} & \frac{1}{4} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 2 & 1 \\ 6 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{4} & 0 \\ -\frac{1}{2} & \frac{1}{4} & 1 \end{bmatrix}$$

$$\begin{array}{ccc} \frac{1}{2} & \frac{3}{4} & 1 \\ -\frac{1}{2} & \frac{1}{2} & 2 \\ \frac{1}{2} & \frac{5}{4} & 0 \end{array}$$

$$r^2 - 6r + 9 = 0$$

$$\text{hom: } 36 - 9(4) = 0$$

$$\frac{-(-6) \pm 0}{2} = 3$$

$$\text{hence: } c_1 e^{3t} + c_2 t e^{3t} + e^t$$

$$\frac{6 - \sqrt{36 - 4(3)(12)}}{6}$$

$$\frac{108}{6} \quad 18$$

$$\frac{6 - \sqrt{108}}{6}$$

$$\frac{18 \cdot 6}{3 \cdot 3 \cdot 3 \cdot 2 \cdot 2}$$

$$\frac{6 - 6i\sqrt{3}}{6} = 1 \pm i\sqrt{3}$$

$$6\sqrt{3}$$

$$6 + 1$$

$$2e^{-\frac{\pi}{3}i}$$

$$p(1) = 7$$

$$(a_4)(z-1-i)(z-1+i)(z-2)^2 = p(z)$$

$$(a_4)(-i)(i)(-1)^2 = 7$$

$$a_4 = 7$$

$$7(z-1-i)(z-1+i)(z-2)^2$$

$$f(3) = 3(2) - 1 = 5$$

$$f(4) = 3(5) - 2 = 13$$

$$f(5) = 3(13) - 5 = 34$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix}^{-1}$$

$$(-t+s)'' + 2(-t+s)' - 8(-t+s) = at+b$$

$$0 + 2(-1) - 8(-t+s) = at+b$$

$$-2 + 8t - 40 = at+b$$

$$-42 + 8t = at+b$$

$$r^2 + 2r - 8 = 0$$

$$(r+4)(r-2)$$

$$\begin{matrix} -4 & 2 \end{matrix}$$

$$\begin{bmatrix} -1 & 4 & 4 \\ 0 & 7 & 8 \\ 0 & -4 & -5 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \checkmark$$

$$\begin{bmatrix} -1 & 4 & 4 \\ 0 & 7 & 8 \\ 0 & -4 & -5 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ -1 \end{bmatrix} \checkmark$$

$$\begin{bmatrix} -1 & 4 & 4 \\ 0 & 7 & 8 \\ 0 & -4 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ -6 \\ -6 \end{bmatrix}$$

$$-14 + 8$$

$$\det(A^T \cdot A^{-1} + A \cdot A \cdot A^{-1}) \\ = \det(A + I)$$

$$\begin{array}{cccc|c} 1 & 1 & 3 & 1 & 7 \\ -1 & 0 & 1 & 0 & -5 \end{array} \rightarrow \begin{bmatrix} 1 & 1 & 3 & 1 & 7 \\ 0 & 1 & 4 & 1 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -1 & 0 & 5 \\ 0 & 1 & 4 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 5 \\ 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -4 \\ 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$z^2 - 2z + 5$$

$$\frac{2 \pm \sqrt{4 - 4(5)}}{2}$$

$$\frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2}$$

$$1 - 2i \text{ or } 1 + 2i$$

$$1 \quad 1 \quad \sqrt{4+1}$$

$$3.14$$

$$\sqrt{2}$$

$$\sigma_2$$

$$\textcircled{4}$$

$$4 + 2$$

$$6$$

$$r^2 - 2r + 5$$

$$\frac{-(-2) \pm \sqrt{4 - 4(5)}}{2}$$

$$\frac{2 \pm 4i}{2} = 1 \pm 2i$$

$$\sim 2 \cos(2t)$$

$$\begin{matrix} 2-\lambda & 0 & 0 \\ 0 & a-\lambda & 1 \\ 0 & -1 & 1-\lambda \end{matrix} \quad (2-\lambda)(6-\lambda(1-\lambda)+1)$$

$$\begin{matrix} 0 & a-\lambda & 1 \\ 0 & -1 & 1-\lambda \end{matrix} \quad (a-\lambda)(a-a\lambda-\lambda+\lambda^2+1)$$

$$\begin{matrix} 2-\lambda & 0 & 0 \\ 0 & a-\lambda & 1 \\ 0 & -1 & 1-\lambda \end{matrix} \quad (2-\lambda)(a-\lambda(a-1)+\lambda^2+1)$$

$$a-2a-2+5$$

$$-a-3=0$$

$$a=3$$

$$a-2(a-1)+4+1=0$$

$$c_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$c_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$c_3 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$4 - 4$$

$$-2 - z^2 + 3 + 3z$$

$$-2z^2 + 3z + 1$$

$$z^3 = -i$$

$$(re^{i\theta})^3 = -i$$

same mod:

$$r^3 = 1$$

$$r = 1$$

same arg

$$3\theta = \arg(-i)$$

$$\theta = \frac{-\pi}{6} + \frac{2k\pi}{3}$$

$$e^{-\frac{\pi i}{6}} \quad e^{i\frac{\pi}{6}} \quad e^{i\frac{\pi}{2}} \quad e^{-\frac{5\pi i}{6}}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 2 & 4 \\ 3 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 4 & -4 \\ 2 & 4 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{rank} = 2$$

$$\text{null}(B) = 1$$

according to Rank Nullity
Theorem

homogenous

$$\begin{bmatrix} f_1'(t) \\ f_2'(t) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} e^t \\ 2e^t \end{bmatrix} \quad \hookrightarrow \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} e^t \\ e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} e^t + e^{2t} \\ 0 & 2e^{2t} \end{bmatrix}$$

$$f_2'(t) = 2f_2(t)$$

$$f(t) = c_2 e^{2t}$$

$$f_1'(t) = f_1(t) + c_2 e^{2t}$$

$$f(t) = e^t \int e^{-t} \cdot c_2 e^{2t}$$

$$= e^t \int e^t c_2$$

$$= e^t \cdot [c_2 e^t + c_1]$$

$$= e^{2t} c_2 + e^t c_1$$

$$\begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ c_2 \end{bmatrix}$$

$$c_2 = 1 \quad c_1 = -1$$

$$\dim(V) = 4$$

$$h) \left[2 \left(\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \right) \right]_{\beta} = {}_{\beta} [L]_{\beta} \cdot [v]_{\beta}$$

$$[v]_{\beta} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$[L(v)]_{\beta} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$L(v) = 4 \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + 2 \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + 1 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}$$

Since the mapping matrix has full column rank, the kernel of the matrix is trivial and zero dimensional. Hence, the matrix is injective.

b) The function is not injective because
split into two cases:

$$x^2 - 3x + 2 - 3(x-1)$$

$$x^2 - 3x + 2 - 3x + 3$$

$$x^2 - 6x + 5$$

$$\text{Case 2: } x^2 - 3x + 2 - 3(-x-1)$$

$$x^2 - 3x + 2 + 3x + 3$$

$$x^2 - 1$$

parabolic



In a parabolic function, different
x-values can produce the same x
value, so not injective

image set:

$$\text{case 1: } y \in \mathbb{R}, y \geq -1$$

$$y \geq -4$$

image set: $[-1, \infty) \cup [-4, \infty)$
 $[-4, \infty)$

$$|2-i| = 2$$

$$|a+bi-i| = 2$$

$$\sqrt{a^2 + (b-1)^2} = 2$$

$$a^2 + (b-1)^2 = (2)^2$$

circle of radius 2

Infinite vertical line

General point on B: $1+ib$

$$(1)^2 + (y-1)^2 = 4$$

$$(y-1)^2 = 3$$

$$y-1 = \pm\sqrt{3}$$

$$y = 1 \pm \sqrt{3}$$

$$(1-i)^{80}$$

$$\left(\sqrt{2} e^{-\pi/4}\right)^{80}$$

$$(\sqrt{2})^{80} e^{-20\pi}$$

$$2^{40} \cos(-20\pi) + i \sin(-20\pi)$$

$$2^{40} (1)$$

$$r_1 e^{i \frac{5\pi}{6}} \times r_2 e^{i \frac{2\pi}{3}}$$

$$e^{i \frac{5\pi}{6} + i \frac{2\pi}{3}} = e^{i \frac{3\pi}{2}}$$

$$e^z = 2 - 2i$$

$$z = a + bi$$

$$e^a \cdot e^{bi} = 2 - 2i$$

2-2i in euler form:

$$r = \sqrt{(2)^2 + (-2)^2} = 2\sqrt{2}$$

$$\text{argument} = \frac{-2}{2} = \tan^{-1}(-1)$$

$$2\sqrt{2} e^{i \frac{-\pi}{4} + 2p\pi} = \frac{-\pi}{4} + 2p\pi$$

$$e^a e^{bi} = 2\sqrt{2} e^{i \frac{-\pi}{4} + 2p\pi}$$

$$e^a = 2\sqrt{2}$$

$$a = \ln(2\sqrt{2})$$

$$e^{bi} = e^{i \frac{-\pi}{4} + 2p\pi}$$

$$b = \frac{-\pi}{4} + 2p\pi$$

$$\text{hence } z = \ln(2\sqrt{2}) + \left(\frac{-\pi}{4} + 2p\pi \right) i$$

$$z_1 = \ln\sqrt{2} - \frac{\pi}{4} i \quad z_2 = \ln\sqrt{2} + \frac{7\pi}{4} i$$

$$\begin{array}{r}
 \underline{z+2} \mid z^4 + 4z^3 + 5z^2 + 4z + 4 \mid \overline{z^3 + 2z^2 + z + 2} \\
 \underline{z^4 + 2z^3} \\
 2z^3 + 5z^2 \\
 \underline{2z^3 + 4z^2} \\
 z^2 + 4z \\
 \underline{z^2 + 2z} \\
 2z + 4 \\
 \underline{2z + 4} \\
 0
 \end{array}$$

$$\mid z^3 + 2z^2 + z + 2$$

Ordered basis:

$$\begin{bmatrix} -1 & 5 & 7 \\ 1 & 4 & 11 \\ 0 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 5 & 7 \\ 0 & 9 & 18 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 5 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

ordered basis: $\{v_1, v_2\}$

$$\rightarrow L(f_1) = f_1' + f_1 - 1$$

$$L(f_2) = f_2' + f_2 - 1$$

$$L(f_1) + L(f_2) = f_1' + f_1 + f_2' + f_2 - 2$$

$$L(f_1 + f_2) = f_1' + f_2' + f_1 + f_2 - 1$$

not linear

$${}_B[F]_Y =$$

$$[F[r_1]]_B = \begin{bmatrix} 1 & 1 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}_B$$

$$[F[r_2]]_B = \begin{bmatrix} 1 & 1 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}_B$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}_\beta = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\begin{array}{rcl} c_1 - 2c_2 & = & 2 \\ 4c_1 + 2c_2 & = & 2 \\ \hline 5c_1 & = & 4 \end{array}$$

$$c_1 = \frac{4}{5}$$

$$c_2 = \frac{c_1 - 2}{2} = \frac{\frac{4}{5} - 2}{2} \quad \begin{array}{l} -3 \\ \cancel{6} \times \frac{1}{2} \\ \frac{5}{2} \end{array}$$

$$\begin{bmatrix} 1 \\ 5 \end{bmatrix}_\beta = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\begin{array}{rcl} c_1 - 2c_2 & = & 1 \\ 4c_1 + 2c_2 & = & 10 \\ \hline 5c_1 & = & 11 \end{array}$$

$$c_1 = \frac{11}{5}$$

$$c_2 = \frac{c_1 - 1}{2} = \frac{\frac{11}{5} - 1}{2}$$

$$3/5$$

$$\begin{array}{l} 3 \\ \cancel{6} \times \frac{1}{2} \\ \frac{5}{2} \end{array}$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ -2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ -2 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 2 \\ -2 \\ 0 \end{bmatrix}$$

$$(2-\lambda)((1-\lambda)(1-\lambda)-1)$$

$$(2-\lambda)(1-2\lambda+\lambda^2-1)$$

$$(2-\lambda)(\lambda^2-2\lambda)$$

$$(2-\lambda)(\lambda)(\lambda-2)$$

$$\lambda=0, \lambda=2$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -1 \\ 4 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$-1 - (-1)$$

$$\lambda = 2 \begin{bmatrix} 0 & 0 & 0 \\ 2 & -1 & -1 \\ 2 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1/2 \\ 0 \\ 1 \end{bmatrix}$$

$$Ax = b$$

$$A(3v_p) = 3Av_p = 3b$$

$$3b = b$$

$$\text{unless } b = 0$$

$$|2x-1| = |-5x+3| \wedge T$$

$$(2x+1 = 5x+3) \vee (2x+1 = -5x+3)$$

$$(4 = 3x) \vee (7x = 2)$$

$$\left(x = \frac{3}{4}\right) \vee \left(x = \frac{2}{7}\right)$$

$$|x| = 2x+1 \wedge T$$

$$|x| = 2x+1 \wedge x < 0 \vee x \geq 0$$

$$(-x = 2x+1 \wedge x < 0) \vee (x = 2x+1 \wedge x \geq 0)$$

$$\begin{array}{l} -3x = 1 \\ x = -\frac{1}{3} \end{array} \vee \underbrace{(-x = 1, x \geq 0)}_F$$

$$\text{hence } x = -\frac{1}{3}$$

$$e^z = 1$$

$$e^z = 1e^{i0}$$

$$\text{principal arg} = 0$$

$$\text{Other args} = (0 + 2\pi i)$$

$$\{i \cdot p \cdot 2\pi \mid p \in \mathbb{Z}\}$$

$$e^z = e$$

$$e^z = e \cdot e^{0 + 2p\pi i} = e^{1 + 2p\pi i}$$

$$z = \{1 + 2p\pi i \mid p \in \mathbb{Z}\}$$

$$e^z = 2i$$

$$e^a \cdot e^{bi} = 2e^{(\frac{\pi}{2} + 2p\pi)i}$$

$$e^a = 2$$

$$a = \ln(2)$$

$$b = \frac{\pi}{2} + 2p\pi$$

$$z = a + bi$$

$$= \left\{ \ln 2 + i \left(\frac{\pi}{2} + 2p\pi \right) \right\}$$

$$e^{2z} = 2 + i$$

$$z = a + bi$$

$$e^{2a} \cdot e^{2bi} = 2 + i$$

$2 + i$ in Euler form

$$\text{modulus: } \sqrt{(2)^2 + (1)^2} = \sqrt{5}$$

$$\text{Argument} = \tan^{-1}\left(\frac{1}{2}\right) + 2\pi p$$

$$e^{2a} \cdot e^{2bi} = \sqrt{5} e^{\tan^{-1}(1/2) + 2\pi p}$$

$$e^{2a} = \sqrt{5}$$

$$2a = \ln \sqrt{5}$$

$$2a = \frac{1}{2} \ln 5$$

$$a = \frac{1}{4} \ln 5$$

$$2b = \tan^{-1}\left(\frac{1}{2}\right) + 2\pi p$$

$$b = \frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right) + \pi p$$

$$\frac{1}{4} \ln 5 + \left(\frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right) + \pi p\right)i$$

$$(e^z)^4 = 1$$

$$w = e^z$$

$$w^4 = 1 e^{0 + 2\pi p}$$

$$w = e^{\frac{\pi}{2} pi}$$

$$p=0 \quad e^0 = 1$$

$$p=1, \quad e^0 = i \quad p=2 \quad e^0 = -1$$

$$e^z = 1$$

$$2\pi i$$